

EXEC_13: Fixed-Scale t_N Analysis (EJ-204)

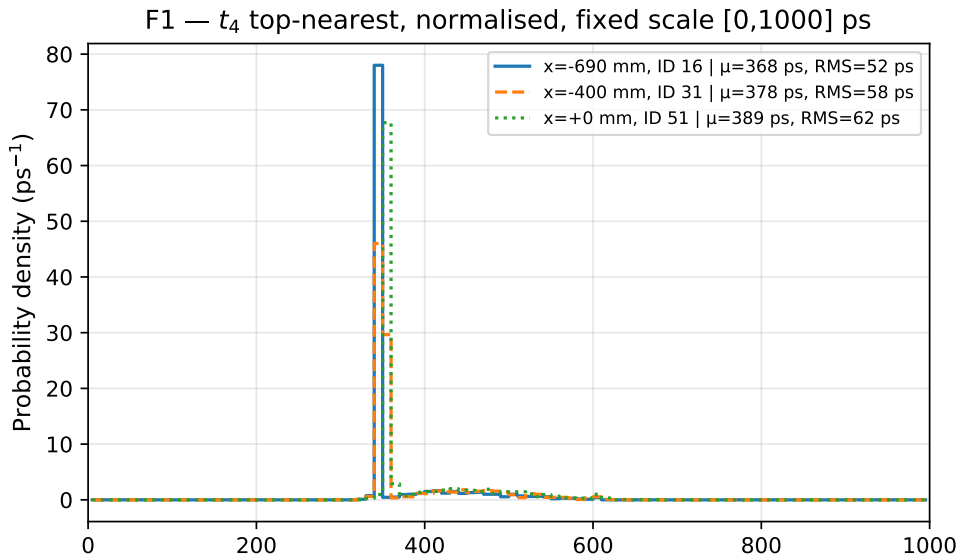
Gerardo's request — meeting 12 Jun 2026

René Ríos

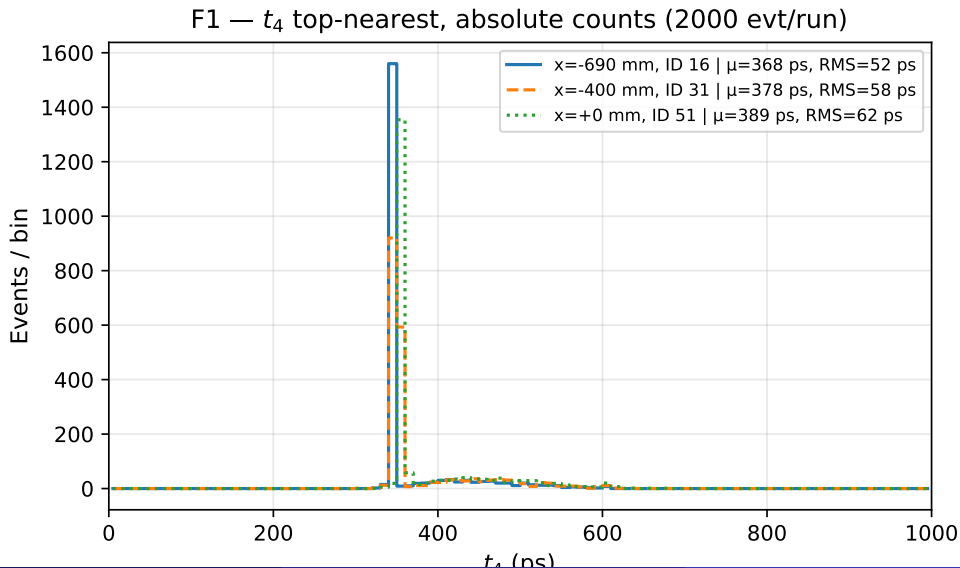
Universidad de La Serena (ULS)

2026-06-13

F1 — Overlay t_4 , normalised (fixed scale)

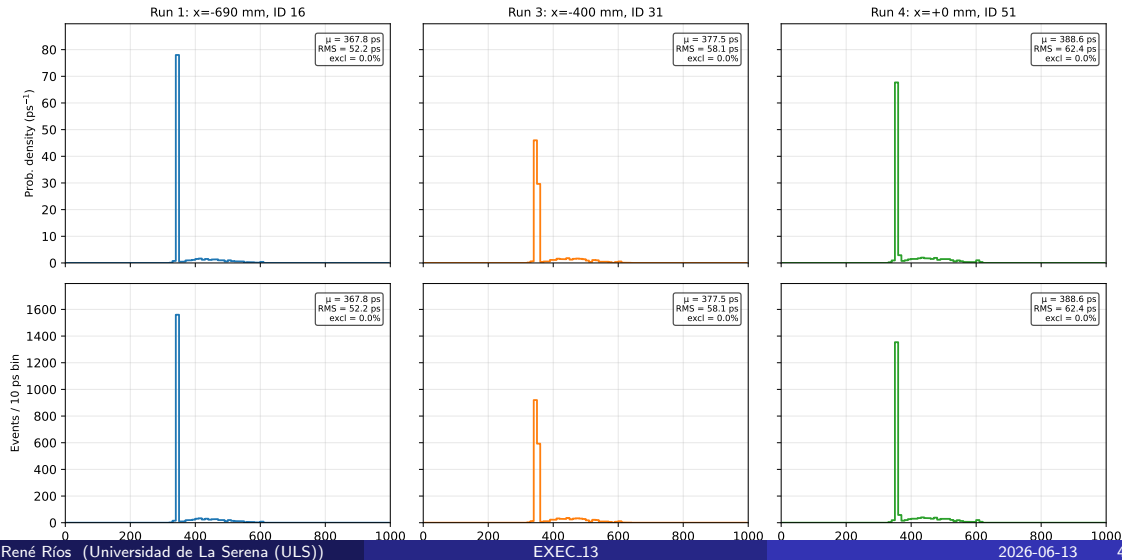


F1 — Overlay t_4 , absolute counts (fixed scale)

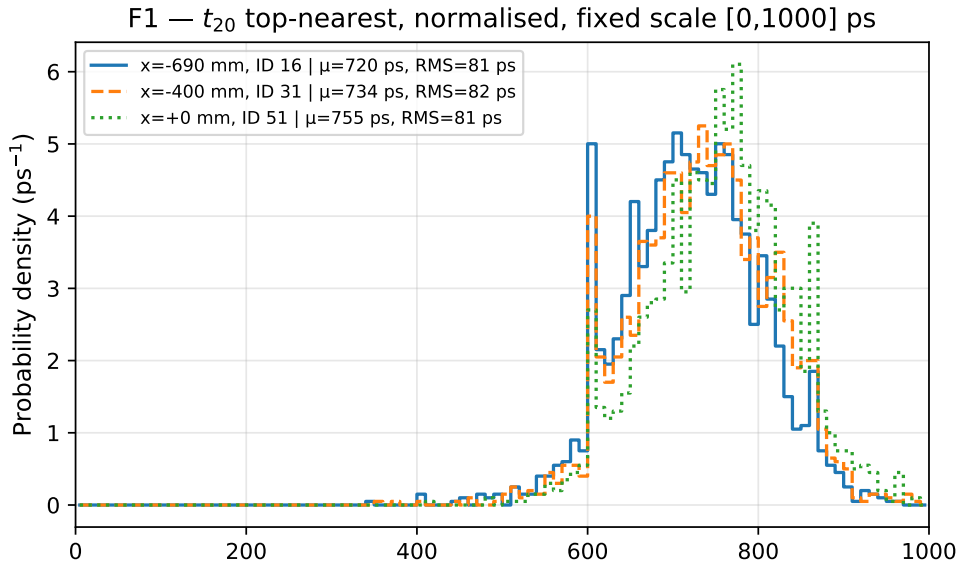


F1 — Individual panels t_4 (identical scale)

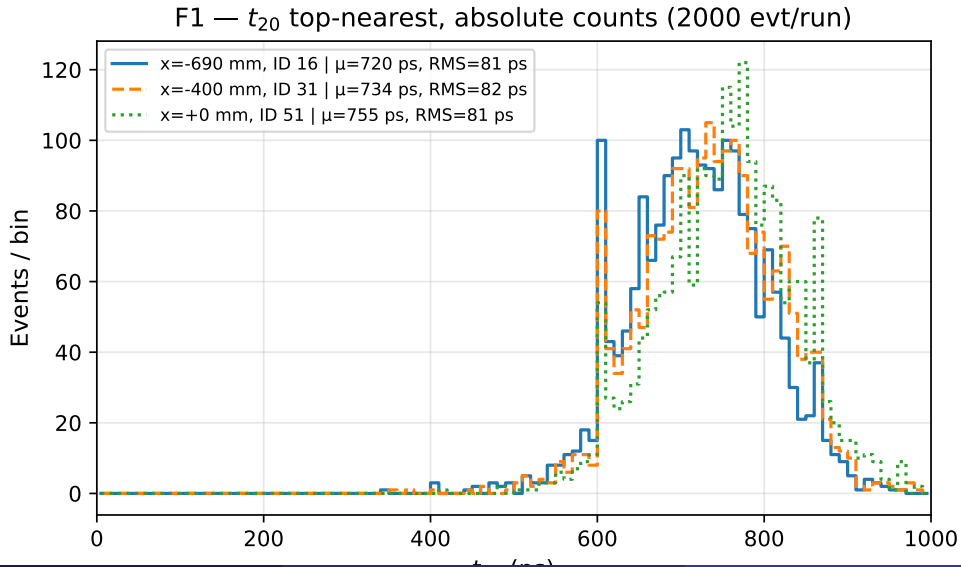
F1 — t_4 top-nearest | 10 ps bins, fixed scale (3-run comparison)



F1 — Overlay t_{20} , normalised (fixed scale)

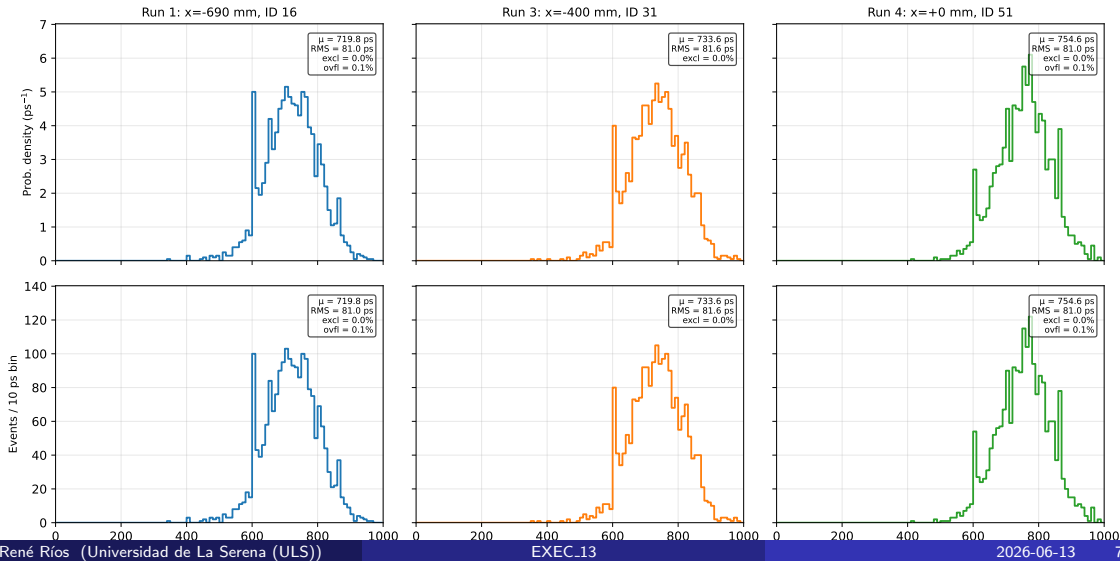


F1 — Overlay t_{20} , absolute counts (fixed scale)



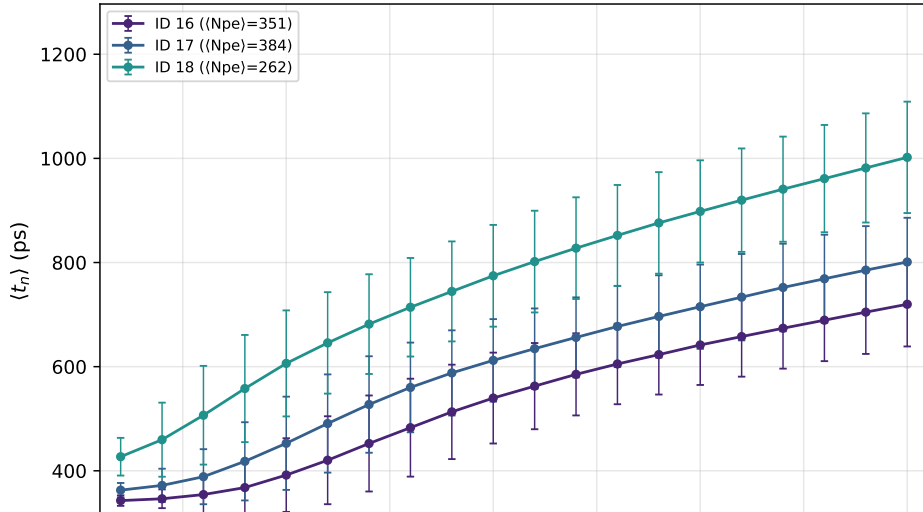
F1 — Individual panels t_{20} (identical scale)

F1 — t_{20} top-nearest | 10 ps bins, fixed scale (3-run comparison)



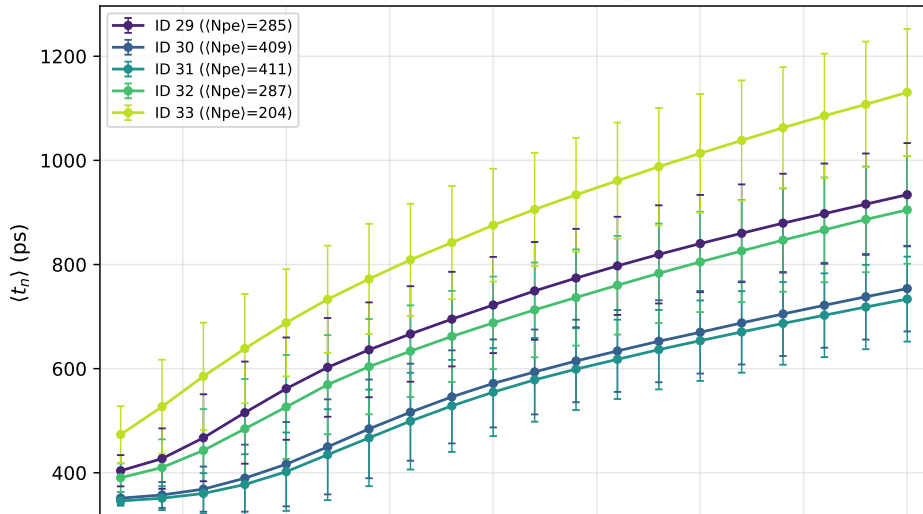
F2 — Run 1: $x = -690$ mm cluster dispersion

F2 — Run 1: $x = -690$ mm, nearest ± 2 cluster
(dashed: stat floor $\sqrt{n} \cdot \tau_d / \langle N_{pe} \rangle$)



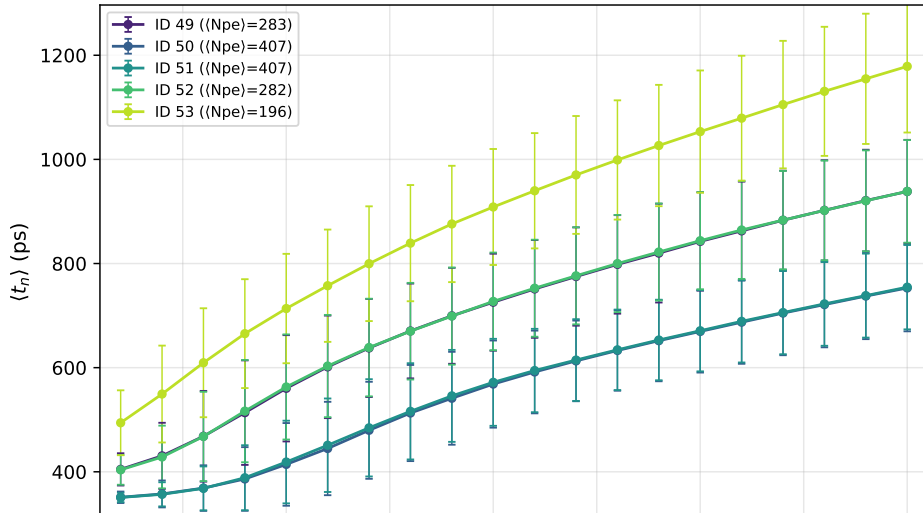
F2 — Run 3: $x = -400$ mm cluster dispersion

F2 — Run 3: $x = -400$ mm, nearest ± 2 cluster
(dashed: stat floor $\sqrt{n} \cdot \tau_d / \langle N_{pe} \rangle$)



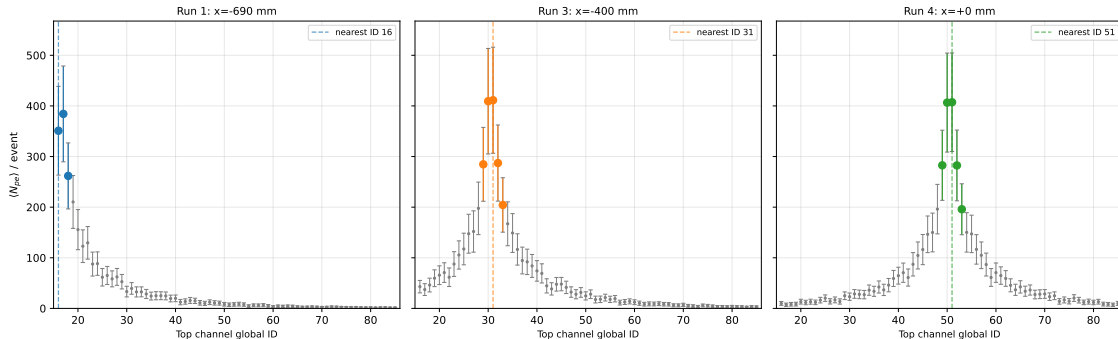
F2 — Run 4: $x = +0$ mm cluster dispersion

F2 — Run 4: $x = +0$ mm, nearest ± 2 cluster
(dashed: stat floor $\sqrt{n} \cdot \tau_d / \langle N_{pe} \rangle$)



F3 — Top N_{pe} profile (all 70 channels, fixed scale)

F3 — Top Npe profile (all 70 channels, fixed scale)

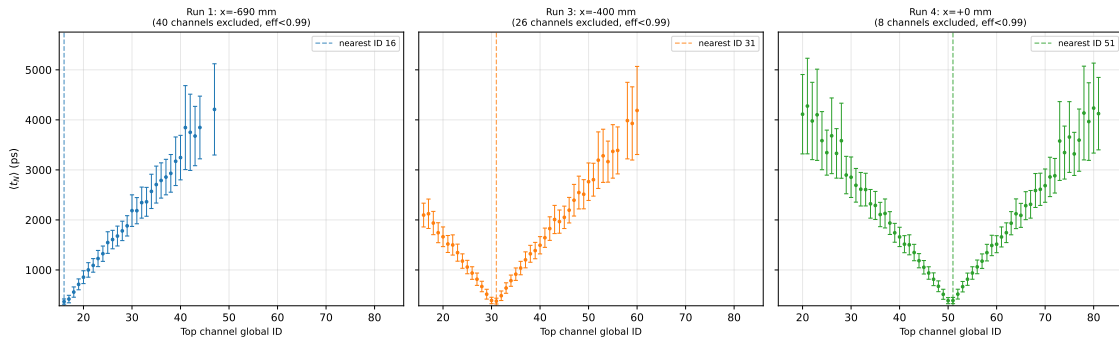


How to read this plot

Why: Verify that the nearest channel is correctly identified as the Npe maximum; shows photon budget across the full top array. **Axes:** x: global channel ID (16–85); y: $\langle N_{pe} \rangle / \text{event} \pm \text{RMS}$. Highlighted: cluster ± 2 . **Takeaway:** Npe falls off rapidly away from the track; ± 2 neighbours still collect useful light.

F4 — Timing profile $\langle t_4 \rangle$ (eff ≥ 0.99 , fixed scale)

F4 — Top timing profile (t_4) (channels with eff ≥ 0.99), fixed scale

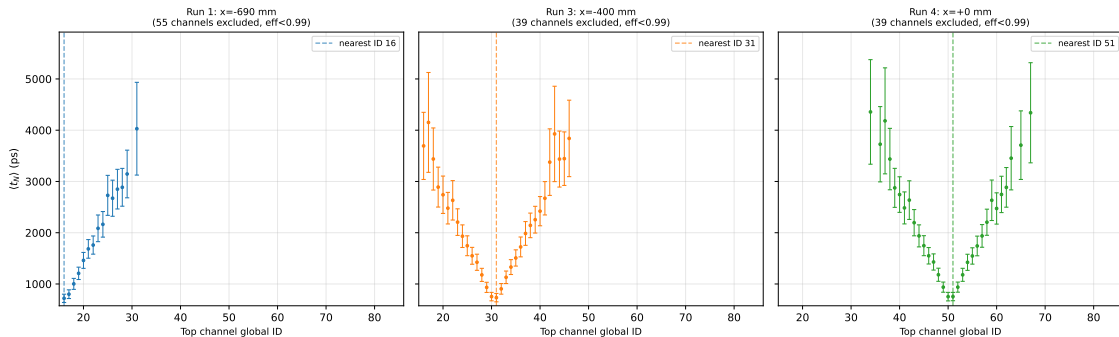


How to read this plot

Why: Only channels with efficiency ≥ 0.99 are shown; far-end channels that never reach threshold are excluded and listed in `exec13_f4_timing_profile.csv`. **Axes:** x: global channel ID; y: $\langle t_N \rangle$ (ps) $\pm$ RMS. **Takeaway:** Timing rises steeply for channels far from the track (longer optical path + lower N_{pe}).

F4 — Timing profile $\langle t_{20} \rangle$ (eff ≥ 0.99 , fixed scale)

F4 — Top timing profile (t_{20}) (channels with eff ≥ 0.99), fixed scale



How to read this plot

Why: Only channels with efficiency ≥ 0.99 are shown; far-end channels that never reach threshold are excluded and listed in `exec13_f4_timing_profile.csv`. **Axes:** x: global channel ID; y: $\langle t_N \rangle$ (ps) $\pm$ RMS. **Takeaway:** Timing rises steeply for channels far from the track (longer optical path + lower N_{pe}).

Backup: validation gates

- **G1:** nearest IDs = $\{-690 \rightarrow 16, -400 \rightarrow 31, 0 \rightarrow 51\}$ (EXEC_12 diagonal) — must pass before plotting
- **G2:** all figure panels share identical axis limits — verified programmatically post-plot
- **G3:** $n_{\text{events}} = 2000$ per run — verified from ROOT file
- **G4:** $n_{\text{in}} + n_{\text{excl}} + n_{\text{ovfl}} = 2000$ per histogram — bookkeeping check

Gate results: `exec13_gates.csv`. Commit: 5783a0d2914b. Stats table: `exec13_f1_tn_histograms.csv` (no manual transcription).

Cierre: método y procedencia

- **Generación:** figuras y CSVs producidos por análisis automático (Claude Code) sobre los ROOT de EXEC_12 (campana EJ-204, OPSC-101). Sin re-simulación.
- **Validación manual pendiente:** R. Ríos verificará en ROOT los valores de μ y RMS de t_4/t_{20} del canal nearest (tolerancia $\sim \text{RMS}/\sqrt{2000}$). Hasta esa validación, los valores exhibidos son los del pipeline del agente.
- **Infraestructura de cómputo:** simulaciones de EXEC_12 corridas en el servidor t0minidaq (solo detalle logístico, no afiliación).
- **Procedencia de commits:** el slide de gates cita 5783a0d (HEAD antes del análisis = tag pre-exec13-checkpoint); el commit que genera las figuras y CSVs es c030e880. El autoritativo de la fuente de código es c030e880; los ROOT de datos viven en \$DATA_ROOT (fuera del repositorio).

Cierre: observaciones y temas abiertos

- **F1 — sub-resolución del núcleo de t_4 :** $\sigma_{\text{fit}}(t_4) \approx 8\text{--}9\text{ ps}$ (EXEC_12); un bin de 10 ps no resuelve el núcleo. Pendiente: bin $\leq 5\text{ ps}$ con zoom, o ajuste gaussiano al core.
- **F1 — artefacto de binning:** diferencias de altura de pico en la versión normalizada de t_4 reflejan en qué borde de bin cae el modo (no anchos físicos distintos). Interpretación cuidadosa requerida.
- **F3 — efecto window-track en $x = -690\text{ mm}$:** nearest geométrico ID 16 ($\langle N_{pe} \rangle = 350.9$) colecta menos que ID 17 ($\langle N_{pe} \rangle = 384.2$). La diagonal de EXEC_12 ancla el nearest en ID 16 (criterio $\text{max-}N_{pe}$, que rompe el empate geométrico). Efecto window-track documentado; no es un error.
- **F2 — rango de n verificado:** $n = 1 \dots 20$ según script (TN_MAX_ORDER=20); confirmado leyendo `exec13_f2_cluster_dispersion.csv`. Consistente con lo especificado.
- **Resultado positivo — t_{20} del nearest:** $\mu \approx 720/734/755\text{ ps}$, $\text{RMS} \approx 81\text{ ps}$ para $x = -690/-400/0\text{ mm}$ (valores exactos del CSV: $719.8/733.6/754.6\text{ ps}$). Timing del Top aproximadamente independiente de la posición de impacto longitudinal.